敏實科技大學
MINTH UNIVERSITY OF SCIENCE AND TECHNOLOGY
人工智慧專業大學

MITUST · Nguyên lý xe tự lái · Tuần 5

Hôm nay đọc bảng thông số thật

今天來讀真正的規格表

Ba thanh trượt hai tuần vừa rồi — hôm nay tìm chúng trên datasheet của hãng

前兩週拉的那三個滑桿，今天在廠商的規格表上把它們找出來

工具｜Công cụ ：
ch-neural.github.io/MITUST-Principles-of-Self-Driving-Cars/week5/spec/

第五週 第 1 節　2026/10/08（四）09:10–10:00　CH Tseng

Mở bàn đọc thông số

打開規格判讀台

https://ch-neural.github.io/MITUST-Principles-of-Self-Driving-Cars/week5/spec/

1. **Bốn model: ba cái là hàng thật, một cái là model ảo để so sánh giá rẻ**
   四個型號：三個是真的產品，一個是便宜的教學用型號，拿來比較
2. **Có nút «chỉ tiếng Anh»: bảng thông số thật thì không có bản dịch**
   有一個「只看英文原文」的按鈕：真的規格表上是不會有翻譯的
3. **Tiết 3 sẽ dùng suốt 50 phút, mỗi người tự làm và nộp**
   第三節整節都會用它，而且是每個人自己做、自己交

Quét mã QR bằng camera điện thoại

用手機相機掃這個 QR

Mở xong bấm thử từng model một lượt cho quen.

打開之後先把四個型號各按一次，熟悉一下畫面。

Phần 1 / 3

Ba thanh trượt của tuần trước, nằm ở đâu trên datasheet

上週那三個滑桿，在規格表的哪一行

Số tia · chiều dài tia · góc quét

射線數· 射線長度· 張角

1 / 2 Ouster OS1-64



Ouster OS1

OPTICAL PERFORMANCE

Range (80% Lambertian reflectivity, 1024 @ 10 Hz mode)	170 m @ >90% detection probability, 100 klx sunlight
Range (10% Lambertian reflectivity, 1024 @ 10 Hz mode)	90 m @ >90% detection probability, 100 klx sunlight
Minimum Range	0.0 m (0.3 m optional, and 0.5 m default)
Vertical Resolution	32, 64, or 128 channels
Horizontal Resolution	512, 1024, or 2048 (configurable)
Rotation Rate	10 or 20 Hz (configurable)
Field of View	Vertical: 42.4° ± 1.0° (+21.2° to -21.2°) Horizontal: 360°

Ouster OS1 datasheet, Rev7 / FW 3.2, p.1 · 節錄 · 讀取 2026/10/02

Lidar quay 360°: rải điểm đều quanh xe

旋轉式光達：點平均灑在車子周圍一整圈

Tầm đo
測距

Range (lấy dòng 10%) = chiều dài tia

Range 看 10% 那行 = 射線長度

Số kênh
線束數

Vertical Resolution = số tia

Vertical Resolution = 射線數

Góc nhìn
視野

Field of View = góc quét, quay cả vòng

Field of View = 張角，它轉一整圈

Ba dòng này chính là ba thanh trượt. Chỉ khác là hãng viết bằng tiếng Anh.

這三行就是那三個滑桿，只是廠商用英文寫。

2 / 2 Hesai AT128P

Hesai AT128 圖：Hesai（hesaitech.com）



Number of channels	128
Instrumented range ①	0.5 to 260 m
Ranging capability	1 to 210 m (at 10% reflectivity)
Ranging accuracy	±5 cm (typical)
Ranging precision	3 cm (1σ)
Horizontal FOV	120°
Horizontal resolution ②	Far-field: **0.1°** (10 Hz), **0.2°** (20 Hz) Near-field: **0.2°** (10 Hz), **0.4°** (20 Hz)
Vertical FOV	25.4° (-12.5° to 12.9°)
Vertical resolution ②	Far-field: **0.2°** Near-field: **0.8°**

② Horizontal and vertical resolution
- Far-field measurement: >7.2 m
- Near-field measurement: 0.5 to 7.2 m

Hesai AT128P User Manual A02, §1.5, p.13 的表 + p.15 的註解② · 節錄 · 讀取 2026/10/02

Lidar cố định hướng trước: dồn điểm vào 120° phía trước

固定式前向光達：把點集中在前方 120 度

Số kênh
線束數

Number of channels = 128

Number of channels = 128

Góc nhìn
視野

Horizontal / Vertical FOV = 120° × 25,4°

Horizontal / Vertical FOV = 120°×25.4°

Phân giải
角解析度

resolution 0,1° / 0,2° — Far-field, 10 Hz

resolution 0.1° / 0.2°——Far-field、10 Hz

Cùng là lidar, một cái rải quanh 360°, một cái dồn vào 120°.
同樣是光達，一個灑滿 360 度，一個集中在 120 度。

Xe máy ở 40 m: OS1 chỉ có 9 điểm, AT128P có tới 121 điểm

40 公尺外的機車：OS1 只有 9 個點，AT128P 有 121 個點

Dòng trên datasheet 規格表上那一行	Ouster OS1-64 Ouster OS1-64	Hesai AT128P Hesai AT128P	Nghĩa là gì 這代表什麼
Góc nhìn ngang 水平視野	360°	120°	Một cái nhìn quanh, một cái nhìn trước 一個看四周，一個只看前面
Phân giải ngang 水平角解析度	0.35°	0.1°	AT128P dày hơn 3 lần theo chiều ngang AT128P 左右密了三倍多
Tầm đo (10%) 測距（10% 反射率）	90 m	210 m	Nhìn xa hơn gấp đôi 看得遠一倍以上
Xe máy ở 40 m 40 公尺的機車	9 điểm	121 điểm	Đây mới là con số quyết định 這才是真正決定用不用得了的數字

Hàng cuối không nằm trên datasheet — nó là do ta tự tính từ ba dòng trên.

最後一行規格表上沒有，是我們自己用上面三行算出來的。

Thông số thứ tư · 第四個規格

Tốc độ cập nhật 10 Hz: mỗi giây chỉ nhìn 10 lần, nên xe luôn biết muộn

更新率 10 Hz ：一秒只看 10 次，車子總是晚一點才知道

Ở 10 Hz, mỗi khung cách nhau 100 ms — chạy 40 km/h là xe đã đi 1,1 m.

10 Hz 的話，每一張畫面隔 100 毫秒——時速 40 的時候，車子已經走了 1.1 公尺。

Minimum Range	0.0 m (0.3 m optional, and 0.5 m default)
Vertical Resolution	32, 64, or 128 channels
Horizontal Resolution	512, 1024, or 2048 (configurable)
Rotation Rate	10 or 20 Hz (configurable)

Ouster OS1 datasheet, Rev7 / FW 3.2, p.1 · 節錄 · 讀取 2026/10/02

Tuần 3 và 4 chưa từng nhắc tới thông số này, nhưng nó quyết định «xe thấy muộn bao nhiêu».

第三、四週從來沒提過這個規格，但它決定了「車子晚多久才知道」。

Xem thử · 點雲到底長什麼樣

Camera tối đen hay cháy trắng, point cloud vẫn thấy xe và cây

相機全黑或一片死白時，點雲還是看得到車和樹



圖：Hesai（hesaitech.com）

Hesai AT128 Lidar Point Cloud View vs Camera View

Hesai Technology（原廠官方頻道） · 0:37（三個情境，播兩次）
https://youtu.be/OkV0Lj-uATE

- Trên là point cloud, dưới là camera cùng lúc — tìm cùng một chiếc xe ở hai bên
  上面是點雲、下面是同一時刻的相機——先在兩邊找出同一台車
- Đường tối ban đêm: camera gần như đen hết, point cloud vẫn thấy cả hàng cây
  夜間暗路：相機幾乎全黑，點雲整排樹都還在
- Ra khỏi hầm: camera cháy trắng cửa hầm, point cloud vẫn thấy rõ mấy chiếc xe
  出隧道：相機出口一片死白，點雲那幾台車看得清清楚楚

Lidar tự mang nguồn sáng. Đó là lý do nó và camera phải đi cùng nhau.
光達自己帶光源。這就是為什麼它和相機要一起裝。

Phần 2 / 3

Tại sao datasheet của camera không có góc nhìn

為什麼相機的規格表上沒有視野

Hai dòng quan trọng nhất lại không nằm trên đó

最重要的兩行，偏偏不在上面

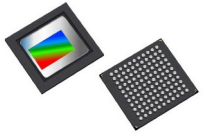
Camera · onsemi AR0233AT / Raspberry Pi Camera Module 3

Datasheet cảm biến ảnh: tìm mãi không thấy góc nhìn

影像感測器的規格表：怎麼找都找不到視野

Chip cảm biến ảnh · onsemi AR0233AT
影像感測器晶片 · onsemi AR0233AT

Specifications



- Optical format – 1/2.5”
- Active pixels – 2048 x 1280
- Pixel size – 3.0 µm
- Color filter array – RGB, RCCG & RCCB
- Interface – MIPI CSI–2, Parallel
- Max dynamic range – 140 dB (4–exp), 120 dB (3–exp)
- Package options – iBGA, CSP, Bare Die

onsemi AR0233AT Product Preview Rev. P2 (11/2017) · 節錄 · 讀取 2026/10/02

Module camera · Raspberry Pi Camera Module 3
相機模組 · Raspberry Pi Camera Module 3

	Camera Module 3	Camera Module 3 NoIR	Camera Module 3 Wide	Camera Module 3 Wide NoIR
Focus range	10cm–∞	10cm–∞	5cm–∞	5cm–∞
Focal length	4.74mm	4.74mm	2.75mm	2.75mm
Diagonal field of view	75 degrees	75 degrees	120 degrees	120 degrees
Horizontal field of view	66 degrees	66 degrees	102 degrees	102 degrees
Vertical field of view	41 degrees	41 degrees	67 degrees	67 degrees

Raspberry Pi Camera Module 3 Product Brief, p.5 · 節錄 · 讀取 2026/10/02

Hai thứ ta cần nhất thì bảng bên trái không hề có. Vì sao?
我們最需要的兩個東西，左邊這張偏偏都沒有。為什麼？

Câu trả lời · 答案



同一幀河內街景：先照各鏡頭的視野取景，再一起框到同一台機車

Góc nhìn nằm ở ống kính, không ở cảm biến

視野不在感測器身上——它在鏡頭身上

Cùng một con chip, lắp ống kính 8 mm thì nhìn được 42°, lắp ống kính 3.6 mm thì nhìn được 81°. Hãng bán chip không biết bạn sẽ lắp ống kính nào, nên họ không thể viết.
同一塊晶片，配 8 mm 的鏡頭看得到 42 度，配 3.6 mm 的鏡頭看得到 81 度。賣晶片的廠商不知道你要配什麼鏡頭，所以寫不出來。

Còn tầm đo: camera không hề đo khoảng cách — nó tính ra, như tuần 3 đã làm.
至於測距：相機根本不量距離，它是算出來的，第三週算過那條式子。

Đọc kỹ · 一行容易讀錯的字

Có dòng ghi «configurable»: đó là chế độ, không phải con số cố định

有些行寫著「可設定」：那是模式，不是固定的數字

Cùng một con lidar, đổi chế độ thì số điểm trên xe máy ở 40 m khác hẳn.
同一顆光達，換一個模式，四十公尺機車身上的點數就整個不一樣。

Lidar Operating Modes	• x 512 @ 10 Hz or 20 Hz • x 1024 @ 10 Hz or 20 Hz • x 2048 @ 10 Hz

Ouster OS1 datasheet, Rev7 / FW 3.2, p.3 · 節錄 · 讀取 2026/10/02

Tiết 3 các bạn sẽ tự đổi: 9 điểm thành 18 điểm, kết luận lật ngược.
第三節各位會自己切： 9 個點變成 18 個點，結論整個翻過來。

Đọc kỹ · 同一行，兩個數字

Range (80% Lambertian reflectivity, 1024 @ 10 Hz mode)	170 m @ >90% detection probability, 100 klx sunlight
Range (10% Lambertian reflectivity, 1024 @ 10 Hz mode)	90 m @ >90% detection probability, 100 klx sunlight

Ouster OS1 datasheet, Rev7 / FW 3.2, p.1 · 節錄 · 讀取 2026/10/02

Xe trắng thấy 170 m, xe đen chỉ 90 m: thiết kế phải dùng 90 m

白車 170 公尺、黑車 90 公尺：設計要用 90 公尺

80%
80%

Vật sáng: xe trắng, tường màu nhạt
淺色、散得回來：白色的車、淺色的牆

10%
10%

Vật tối: xe đen, áo mưa sẫm màu
深色、反射弱：黑色的車、深色雨衣

Dùng số nào
要用哪個

Thiết kế dùng 90 m — ngoài đường có xe đen.
設計用 90 公尺。路上一定有黑色的車。

Quảng cáo hay ghi 170 m. Con số phải dùng để thiết kế là 90 m.
廣告上會寫 170 公尺。真正設計的時候要用的是 90 公尺。

Lên máy · 上機（第一輪）

Mỗi người tự làm: chọn OS1, tìm bốn con số, bấm kiểm tra.

每個人自己做：選 OS1 ，把四個數字找出來，按檢查。

1. **Bốn câu: góc nhìn dọc · tầm đo với vật 10% · phân giải · tốc độ cập nhật**
   四題：垂直視野· 對 10% 反射率的測距· 解析度· 更新率

2. **Sai thì công cụ nói sai ở đâu — đọc xong sửa lại, đừng đoán bừa**
   選錯的話工具會告訴你錯在哪——看完再改，不要亂猜

3. **Làm xong OS1 thì đổi sang AT128P làm lại một lượt**
   OS1 做完，換 AT128P 再做一次

Hai câu của camera sẽ khiến nhiều người bực mình. Đó là chủ ý.
相機那兩題會讓很多人火大。那是故意的。

Tiết 1 · 第一節小結

Mang ba câu này sang tiết 2

帶三句話進第二節

Ba thanh trượt tuần trước = ba dòng trên datasheet: số kênh, góc nhìn, tầm đo. Cộng thêm dòng thứ tư: tốc độ cập nhật.

上週三個滑桿就是規格表上三行：線束數、視野、測距。再加第四行：更新率。

Datasheet cảm biến ảnh không có góc nhìn và tầm đo — góc nhìn do ống kính, khoảng cách là tính ra.

影像感測器的規格表沒有視野和測距——視野看鏡頭，距離是算出來的。

Tầm đo luôn phải hỏi: đo với vật phản xạ bao nhiêu phần trăm. 170 m hay 90 m là hai thế giới khác nhau.

看到測距一定要追問：對百分之幾反射率量的。170 公尺和 90 公尺是兩個世界。

Phụ lục · 附錄（不計時）

Ouster OS1 — toàn bộ datasheet gốc (3 trang thông số)

Ouster OS1 — 廠商規格書原頁（規格那三頁）

OUSTER

Ouster, Inc.
350 Treat Ave.
San Francisco, CA 94110

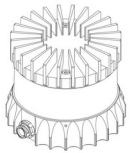
OS1

Mid-Range High-Resolution Imaging Lidar

FIRMWARE VERSION: 3.2.x
HARDWARE VERSION: REV7.0

SUMMARY

The mid-range OS1 lidar sensor features 90 m range on a dark 10% target, a 42.4° vertical field of view, and high reliability for the most rugged conditions. The OS1 is designed for all-weather environments and use in industrial automation, autonomous vehicles, mapping, smart infrastructure, and robotics.

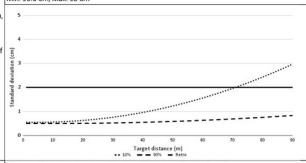
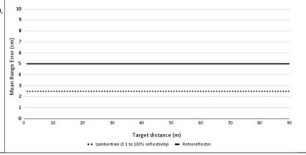


HIGHLIGHTS

- Configurable Minimum Range and Return Ordering
- Low Data Rate Profile now available with Dual Returns
- Camera-grade near-infrared and signal data
- Multi-sensor crosstalk suppression
- Ouster Studio for pointcloud evaluation
- Ouster SDK, ROS, and C++ drivers for SW development

OPTICAL PERFORMANCE

Range (80% Lambertian reflectivity, 1024 @ 10 Hz mode)	170 m @ >90% detection probability, 100 klx sunlight
Range (10% Lambertian reflectivity, 1024 @ 10 Hz mode)	90 m @ >90% detection probability, 100 klx sunlight
Minimum Range	0.0 m (0.3 m optional, and 0.5 m default)
Vertical Resolution	32, 64, or 128 channels
Horizontal Resolution	512, 1024, or 2048 (configurable)
Rotation Rate	10 or 20 Hz (configurable)
Field of View	Vertical: 42.4° ± 1.0° (+21.2° to -21.2°) Horizontal: 360°
Angular Sampling Accuracy	Vertical: ±0.01° / Horizontal: ±0.01°
False Positive Rate	1/10,000
Range Resolution	0.1 cm **Note:** For *Low Data Rate Profile* the Range Resolution = 0.8 cm
# of Returns	up to 2
Return Order	Strongest to Weakest, Farthest to Nearest, and Nearest to Farthest

Range Precision (Typical on Lambertian and Retroreflective targets beyond 1 m, 1024 @ 10 Hz mode, 1 standard deviation) **Note:** Precision is calculated based on the standard deviation of 100 measurements on a static target at a given range	Min: ±0.5 cm, Max: ±3 cm 
Range Accuracy (Typical on Lambertian and Retroreflective targets beyond 1 m, 1024 @ 10 Hz mode) **Note:** Accuracy is calculated based on the error between the mean of 100 measurements on a static target at a given range and the true range	±2.5 cm for lambertian targets, ±5 cm for retroreflective targets 

LASER

Laser Product Class	Class 1 eye-safe per IEC/EN 60825-1: 2014
Laser Wavelength	865 nm
Beam Diameter Exiting Sensor	9.5 mm
Beam Divergence	0.18° (FWHM)

LIDAR OUTPUT

Connection	UDP over gigabit Ethernet
Points Per Second	1,310,720 (32 channel) 2,621,440 (64 channel) 5,242,880 (128 channel)
Data Rate (megabits per second) (Low Data Rate Profile, 1 return, 1024 @ 10 Hz mode)	up to 11.83 Mbps (32 channel) up to 22.32 Mbps (64 channel) up to 43.29 Mbps (128 channel)
Data Rate (megabits per second) (Low Data Rate Profile, 2 returns, 1024 @ 10 Hz mode)	up to 22.32 Mbps (32 channel) up to 43.29 Mbps (64 channel) up to 85.24 Mbps (128 channel)

Data Rate (megabits per second) (Single Return Profile, 1024 @ 10 Hz mode)	up to 32.81 Mbps (32 channel) up to 64.26 Mbps (64 channel) up to 127.18 Mbps (128 channel)
Data Rate (megabits per second) (Dual Return Profile, 1024 @ 10 Hz mode)	up to 43.29 Mbps (32 channel) up to 85.24 Mbps (64 channel) up to 169.12 Mbps (128 channel)
Data Per Point	Range, Signal, Reflectivity, Near-infrared, Channel, Azimuth angle, and Timestamp
Timestamp Resolution	< 1 µs
Data Latency	< 10 ms
Data Integrity	End to End CRC that covers entire data packet

IMU OUTPUT

Connection	UDP over 1000Base-T or 100Base-T1
Samples Per Second	640
Data Per Sample	3 axis gyro, 3 axis accelerometer
Timestamp Resolution	< 1 µs
Data Latency	< 10 ms
Additional Details	InvenSense IAM-20680HT; datasheet for more details: https://invensense.tdk.com/download-pdf/iam-20680ht-datasheet/

CONTROL INTERFACE

Connection	HTTP API
Time Synchronization	Input sources: • IEEE1588 Precision Time Protocol (PTP); Accuracy: <1 ms error • gPTP; Accuracy: <1 ms error • NMEA $GPRMC UART message support • External PPS; Accuracy: <1 ms error • Internal 10 ppm drift clock; Accuracy: <20 ppm error Output sources: • Configurable 1 - 60 Hz output pulse
Lidar Operating Modes	• x 512 @ 10 Hz or 20 Hz • x 1024 @ 10 Hz or 20 Hz • x 2048 @ 10 Hz
Additional Programmability	• Multi-sensor phase lock • Queryable intrinsic calibration information: • Beam angles • IMU pose correction matrix • Return ordering • Minimum range • Azimuth masking • Low-power standby mode

MECHANICAL/ELECTRICAL

Power Consumption	14 - 20 W • 16 W nominal • 28 W peak at startup if operating at -40 °C **Note:** Ouster recommends use of a power supply of no less than 30 W if using in cold conditions
Connector	Standard 100BASE-T or Automotive Standard 1000BASE-T1
Operating Voltage	9.5 V - 51 V • Suitable for 12 VDC to 24 VDC nominal systems • Not suitable for 48 V nominal battery based systems • Under-voltage WARNING level alert occurs at 9.5 VDC at the connector • Under-voltage ERROR level alert occurs at 9.0 VDC at the connector • Below 9.0 VDC at connector, sensor may shutdown • Over-voltage conditions/alarms occur at 51 VDC at the connector • Over-voltage lockout onset at 58 VDC (±1 V) at the connector • Over-voltage lockout release at 55 VDC (±1 V) at the connector

Ouster OS1 datasheet, hardware Rev7 / firmware 3.2 · 官方 PDF 第 1–3 頁原頁全文 · 讀取 2026/10/02 · 版權歸 Ouster ，課堂引用

Phụ lục · 附錄（不計時）

Hesai AT128P — nguyên trang sổ tay (phần thông số)

Hesai AT128P — 使用手冊原頁（規格那兩頁）

1.5. Specifications

SENSOR

Scanning method	1-D rotating mirror (hybrid solid-state)
Number of channels	128
Instrumented range ①	0.5 to 260 m
Ranging capability	1 to 210 m (at 10% reflectivity)
Ranging accuracy	±5 cm (typical)
Ranging precision	3 cm (1σ)
Horizontal FOV	120°
Horizontal resolution ②	Far-field: 0.1° (10 Hz), 0.2° (20 Hz) Near-field: 0.2° (10 Hz), 0.4° (20 Hz)
Vertical FOV	25.4° (-12.5° to 12.9°)
Vertical resolution ②	Far-field: 0.2° Near-field: 0.8°
Frame rate	10/20 Hz
Return mode	Single Return: Last/Strongest/First Dual Return: Last and Strongest, Strongest and First

MECHANICAL/ELECTRICAL/OPERATIONAL

Wavelength	905 nm
Laser class	Class 1 Eye Safe
Ingress protection	IP6K7 & IP6K9K

Notes to specifications

① Instrumented range
- Actual measurement range, confined by the allocated Time of Flight (ToF) for each channel.
- For the point cloud data between 0.5 m and 1 m, ranging accuracy and precision cannot be guaranteed.

② Horizontal and vertical resolution
- Far-field measurement: >7.2 m
- Near-field measurement: 0.5 to 7.2 m

③ Dimensions and weight
- May be different for customized models.
- The mechanical drawings and data exclusively provided for customized models shall prevail.

④ Rated voltage range

Nominal voltages: 12 V and 24 V

⑤ Power consumption
- Typical value, not including accessories such as the connection box.
- The external power supply should be able to provide at least 45 W.

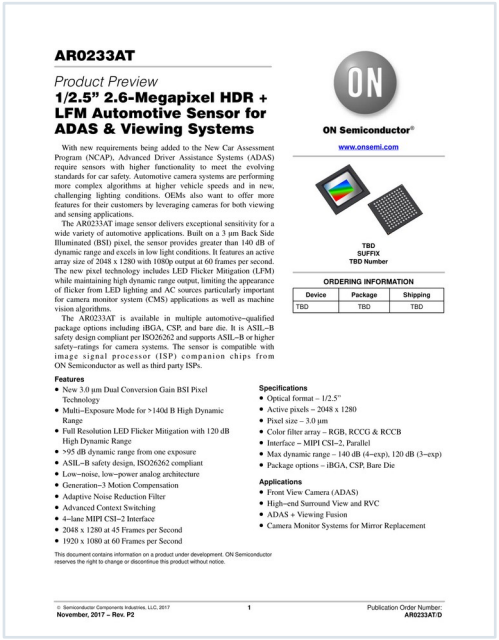
⑥ Valid point rate
- Defined as the point cloud data points (number of returns) generated per second.
- In **Single Return Mode**:
  - Given: horizontal FOV (120°), horizontal resolution (0.1° at 10 Hz), number of channels (128), and frame rate (10 Hz).
  - At 10 Hz, each channel generates 120/0.1 = 1200 points per frame; all channels generate 1200 × 128 = 153 600 points per frame, and 1 536 000 points (10 frames) per second.
- In **Dual Return Mode**, the point rate is twice that in **Single Return Mode**.
- The actual horizontal FOV of each mirror surface is slightly larger than 120°, so the actual point rate is slightly more than the valid point rate.

Hesai AT128P User Manual A02（ en-250410 ）· §1.5 Specifications 第 13 頁與註解第 15 頁原頁全文 · 讀取 2026/10/02 · 版權歸 Hesai ，課堂引用

Phụ lục · 附錄（不計時）

onsemi AR0233AT — toàn bộ datasheet gốc (2 trang)

onsemi AR0233AT — 廠商規格書原頁（全兩頁）



AR0233AT

Product Preview

1/2.5" 2.6-Megapixel HDR + LFM Automotive Sensor for ADAS & Viewing Systems

ON Semiconductor®

www.onsemi.com

TBD
SUFFIX
TBD Number

ORDERING INFORMATION

Device	Package	Shipping
TBD	TBD	TBD

With new requirements being added to the New Car Assessment Program (NCAP), Advanced Driver Assistance Systems (ADAS) require sensors with higher functionality to meet the evolving standards for car safety. Automotive camera systems are performing more complex algorithms at higher vehicle speeds and in new, challenging lighting conditions. OEMs also want to offer more features for their customers by leveraging cameras for both viewing and sensing applications.

The AR0233AT image sensor delivers exceptional sensitivity for a wide variety of automotive applications. Built on a 3 μm Back Side Illuminated (BSI) pixel, the sensor provides greater than 140 dB of dynamic range and excels in low light conditions. It features an active array size of 2048 x 1280 with 1080p output at 60 frames per second. The new pixel technology includes LED Flicker Mitigation (LFM) while maintaining high dynamic range output, limiting the appearance of flicker from LED lighting and AC sources particularly important for camera monitor system (CMS) applications as well as machine vision algorithms.

The AR0233AT is available in multiple automotive−qualified package options including iBGA, CSP, and bare die. It is ASIL−B safety design compliant per ISO26262 and supports ASIL−B or higher safety−ratings for camera systems. The sensor is compatible with image signal processor (ISP) companion chips from ON Semiconductor as well as third party ISPs.

Features

- New 3.0 μm Dual Conversion Gain BSI Pixel Technology
- Multi−Exposure Mode for >140d B High Dynamic Range
- Full Resolution LED Flicker Mitigation with 120 dB High Dynamic Range
- >95 dB dynamic range from one exposure
- ASIL−B safety design, ISO26262 compliant
- Low−noise, low−power analog architecture
- Generation−3 Motion Compensation
- Adaptive Noise Reduction Filter
- Advanced Context Switching
- 4−lane MIPI CSI−2 Interface
- 2048 x 1280 at 45 Frames per Second
- 1920 x 1080 at 60 Frames per Second

Specifications

- Optical format − 1/2.5"
- Active pixels − 2048 x 1280
- Pixel size − 3.0 μm
- Color filter array − RGB, RCCG & RCCB
- Interface − MIPI CSI−2, Parallel
- Max dynamic range − 140 dB (4−exp), 120 dB (3−exp)
- Package options − iBGA, CSP, Bare Die

Applications

- Front View Camera (ADAS)
- High−end Surround View and RVC
- ADAS + Viewing Fusion
- Camera Monitor Systems for Mirror Replacement

This document contains information on a product under development. ON Semiconductor reserves the right to change or discontinue this product without notice.

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Publication Order Number:
AR0233AT/D



AR0233AT

PACKAGE DIMENSIONS

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